Digital Signal Processing

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Other FFT algorithms for time-based extraction

- Principle of FFT algorithm for time-based extraction
- Flow chart of FFT algorithm based on 4 time extraction
- Principle of FFT algorithm based on 3 time extraction
- Flow chart of FFT algorithm based on 3 time extraction

Other FFT algorithms for time-based extraction

The radix-4 time extraction FFT algorithm decomposes a sequence $x[k]$ of length N into four sets of short sequences

$$x_1[k]=x[4k] \qquad k=0,1,\cdots,N/4-1 \qquad N=4^M$$

$$x_2[k]=x[4k+1] \qquad k=0,1,\cdots,N/4-1$$

$$x_3[k]=x[4k+2] \qquad k=0,1,\cdots,N/4-1$$

$$x_4[k]=x[4k+3] \qquad k=0,1,\cdots,N/4-1$$

Calculate the DFT of 4 sets of short sequences separately, and then combine them into the DFT of long sequences.

Short sequence DFT combined with long sequence DFT

$$X[m] = X_1[m] + W_N^m X_2[m] + W_N^{2m} X_3[m] + W_N^{3m} X_4[m]$$

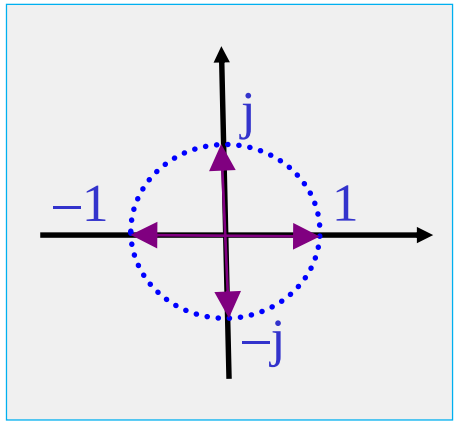
$$X[m+N/4] = X_1[m] + W_4^1 W_N^m X_2[m] + W_4^2 W_N^{2m} X_3[m] + W_4^3 W_N^{3m} X_4[m]$$

$$X[m+2N/4] = X_1[m] + W_4^2 W_N^m X_2[m] + W_4^4 W_N^{2m} X_3[m] + W_4^6 W_N^{3m} X_4[m]$$

$$X[m+3N/4] = X_1[m] + W_4^3 W_N^m X_2[m] + W_4^6 W_N^{2m} X_3[m] + W_4^9 W_N^{3m} X_4[m]$$

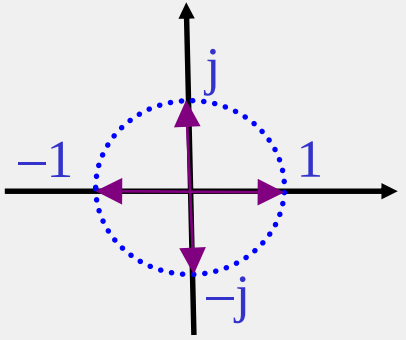
Short sequence DFT combined with long sequence DFT

$$\begin{bmatrix} X[m] \\ X[m+N/4] \\ X[m+2N/4] \\ X[m+3N/4] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^m & 0 & 0 \\ 0 & 0 & W_N^{2m} & 0 \\ 0 & 0 & 0 & W_N^{3m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \\ X_4[m] \end{bmatrix}$$



4-point sequence DFT - Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$



Principle of FFT algorithm for time-based extraction

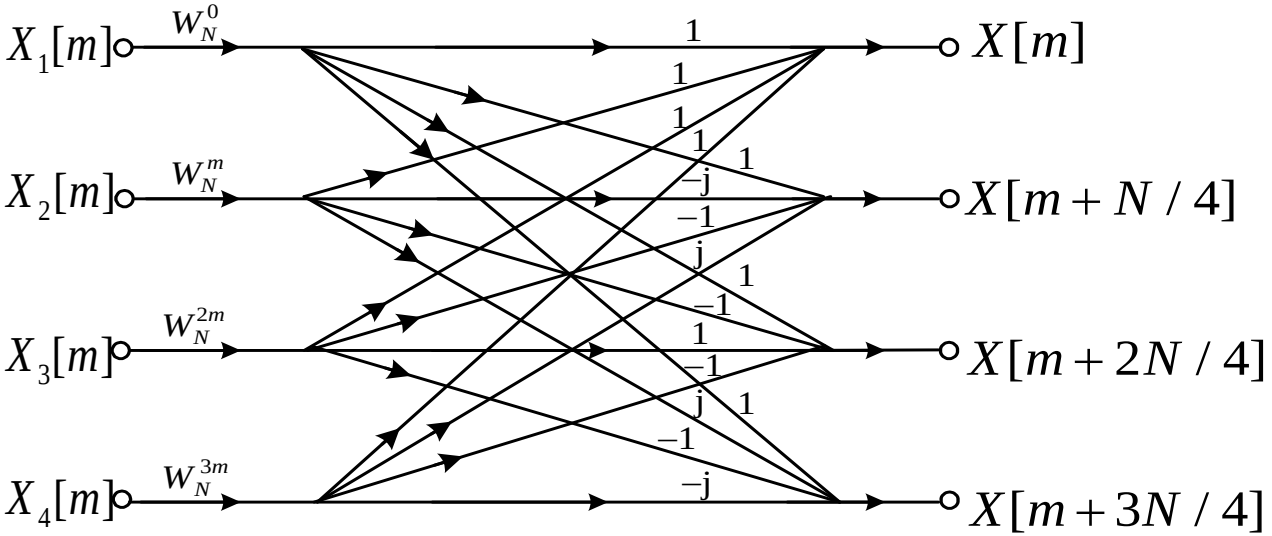
Time domain to frequency domain

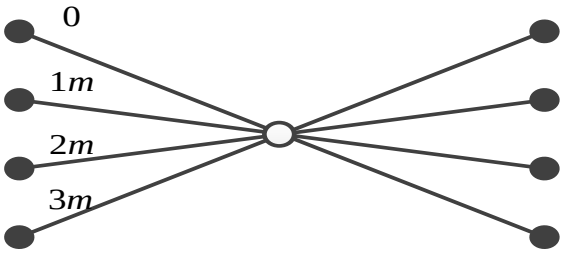
$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Frequency domain synthesis

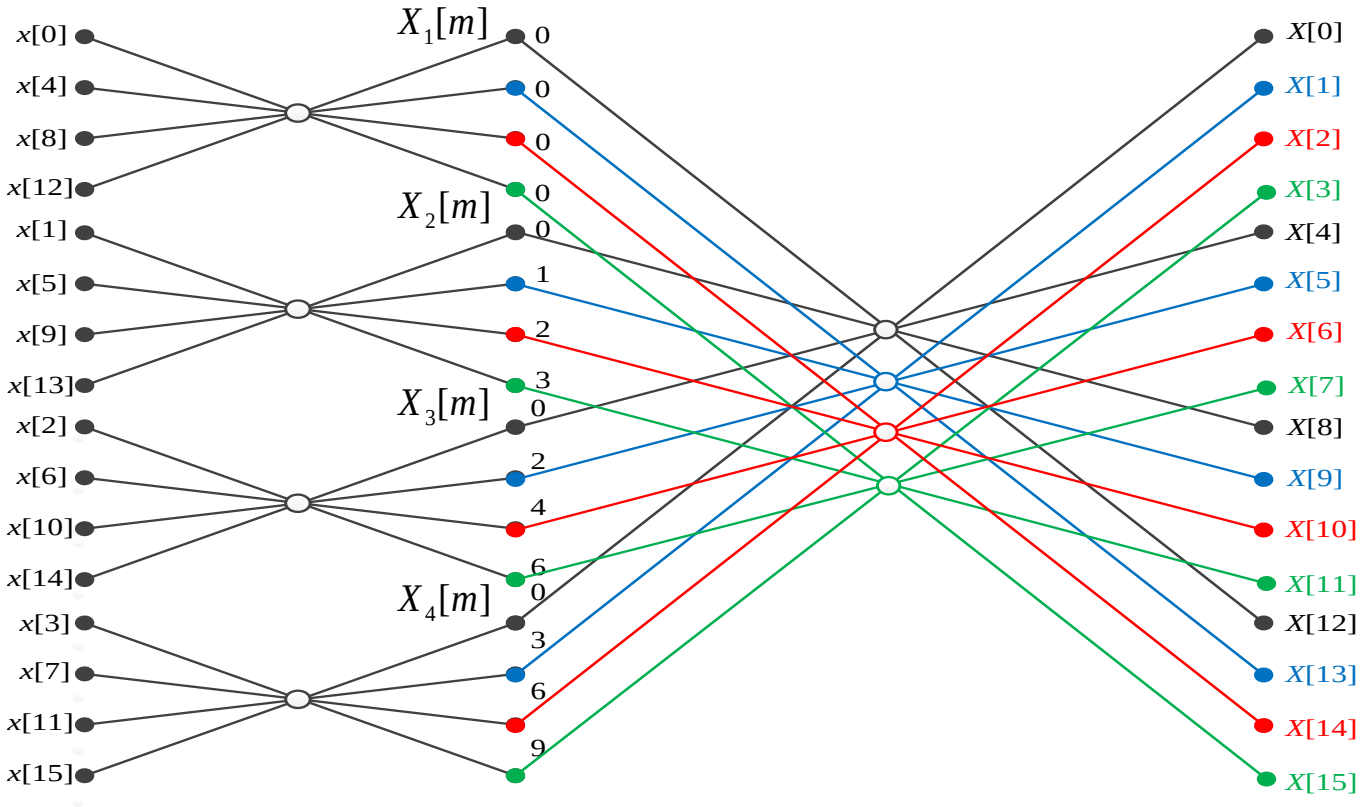
$$\begin{bmatrix} X[m] \\ X[m+N/4] \\ X[m+2N/4] \\ X[m+3N/4] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & \mathrm{j} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{j} & -1 & -\mathrm{j} \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 & 0 \\ 0 & W_N^m & 0 & 0 \\ 0 & 0 & W_N^{2m} & 0 \\ 0 & 0 & 0 & W_N^{3m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \\ X_4[m] \end{bmatrix}$$

The basic butterfly shape in the radix-4 FFT algorithm





Flow chart of FFT algorithm based on 4 time extraction



Flow diagram of 16 point FFT algorithm based on 4-FFT

The radix-3 time extraction FFT algorithm decomposes a sequence $x[k]$ of length N into three sets of short sequences

$$x_1[k]=x[3k] \qquad k=0,1,\cdots,N/3-1$$

$$x_2[k]=x[3k+1] \qquad k=0,1,\cdots,N/3-1$$

$$x_3[k]=x[3k+2] \qquad k=0,1,\cdots,N/3-1$$

$N=3^M$

Calculate the DFT of three sets of short sequences separately, and then combine them into the DFT of long sequences.

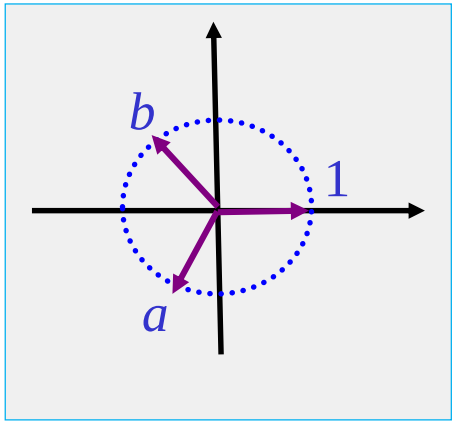
Short sequence DFT combined with long sequence DFT

$$X[m]=X_1[m]+W_N^m X_2[m]+W_N^{2m} X_3[m]$$

$$X[m+N/3]=X_1[m]+W_3^1 W_N^m X_2[m]+W_3^2 W_N^{2m} X_3[m]$$

$$X[m+2N/3]=X_1[m]+W_3^2 W_N^m X_2[m]+W_3^4 W_N^{2m} X_3[m]$$

$$\begin{bmatrix} X[m] \\ X[m+N/3] \\ X[m+2N/3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & b & a \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 \\ 0 & W_N^m & 0 \\ 0 & 0 & W_N^{2m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \end{bmatrix}$$



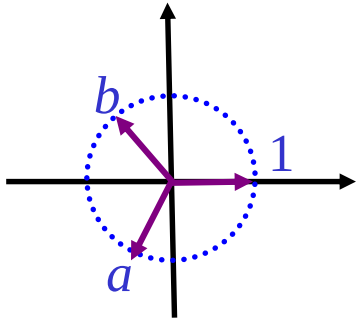
$$a=-\frac{1}{2}-\mathrm{j}\frac{\sqrt{3}}{2}$$

$$b=-\frac{1}{2}+\mathrm{j}\frac{\sqrt{3}}{2}$$

Principle of FFT algorithm based on 3 time extraction

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & b & a \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \end{bmatrix}$$



Frequency domain synthesis

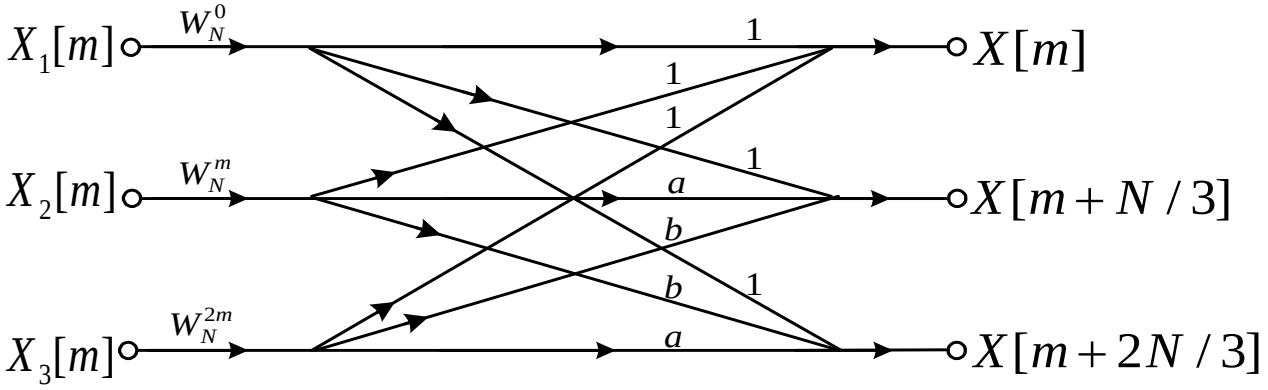
$$\begin{bmatrix} X[m] \\ X[m+N/3] \\ X[m+2N/3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & b & a \end{bmatrix} \begin{bmatrix} W_N^0 & 0 & 0 \\ 0 & W_N^m & 0 \\ 0 & 0 & W_N^{2m} \end{bmatrix} \begin{bmatrix} X_1[m] \\ X_2[m] \\ X_3[m] \end{bmatrix}$$

Flow chart of FFT algorithm based on 3 time extraction

$$X[m]=X_1[m]+W_N^m X_2[m]+W_N^{2m} X_3[m]$$

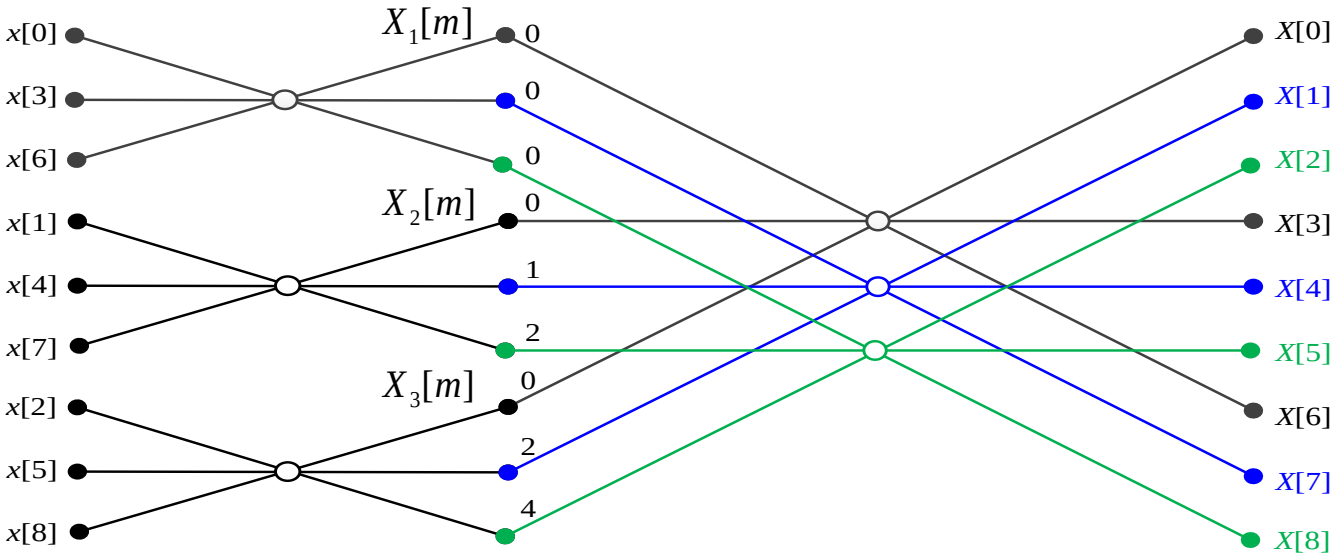
$$X[m+N/3]=X_1[m]+W_3^1 W_N^m X_2[m]+W_3^2 W_N^{2m} X_3[m]$$

$$X[m+2N/3]=X_1[m]+W_3^2 W_N^m X_2[m]+W_3^4 W_N^{2m} X_3[m]$$



Butterfly diagram

Flow chart of FFT algorithm based on 3 time extraction



9-point basis 3-FFT algorithm flow chart

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.